Chem 150 **Answer Key Midterm** **18 February 2013**

Part 1 (6 marks) **Multiple Choice Section**

For each statement below circle the letter that corresponds to the best answer. There is only one correct answer per question. **Each question answered correctly is worth 2 marks**.

1. Find the false statement.

a) Hydrogen bonding is a type of intermolecular interaction.

b) Hydrogen bonding is a particular strong type of dipole-dipole interaction.

c) There are more hydrogen bonds in liquid water than in solid water.

d) Two hydrogen fluoride molecules can hydrogen bond with each other.

e) Hydrogen bonding energies are around 15 and 40 kJ/mol.

2. The osmotic pressure of a 0.129 M magnesium chloride solution at 25°C expressed with three significant figures is:

a) 320 kPa

b) 639 kPa

c) 959 kPa $MgCl_2(s) \leftrightarrow Mg^{2+}(aq) + 2Cl^-(aq)$ thus the molarity of solute particles is 0.387 M

d) Approximately zero since magnesium chloride is not volatile.

e) None of the above answers is correct

3. The Joule Thomson effect explains why the expansion of most gasses is accompanied with

a) a decrease in temperature of the expanding gas.

b) an increase in temperature of the expanding gas.

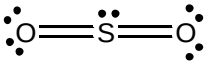
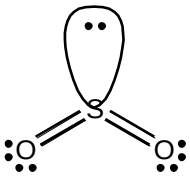
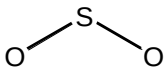
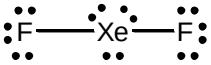
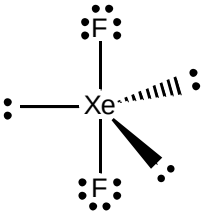
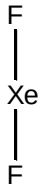
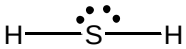
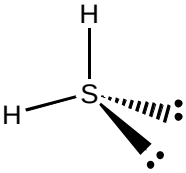
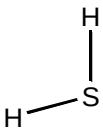
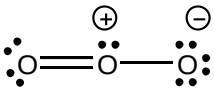
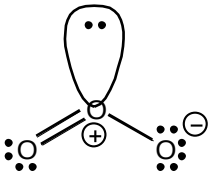
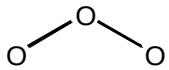
c) no net temperature change of the expanding gas.

d) a reduction in pressure of the expanding gas.

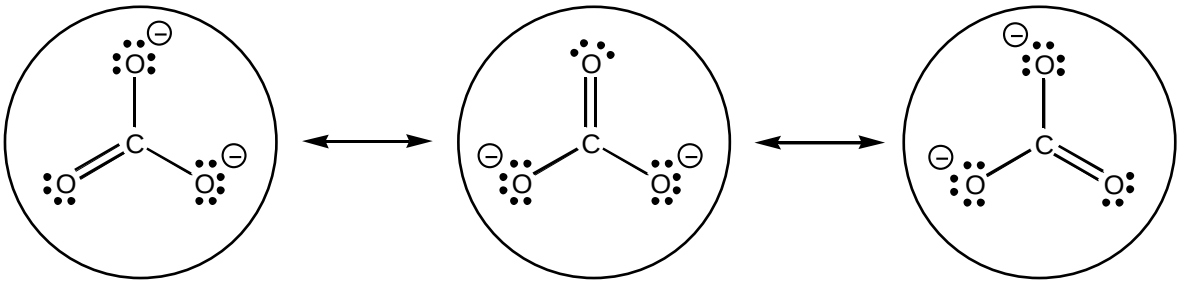
c) none of the above effects are explained by the Joule Thomson effect.

Part 2 Lewis structures, VSEPR and periodic properties

1. (**10.5 marks**) Identify the electron group geometry (EGG), the parent shape and the molecular shape of the species below. Start by writing a valid Lewis structure. Be sure to show all electrons and indicate any non zero formal charges. Continue by drawing the parent shape placing lone pairs and bonding pairs in the appropriate positions. **Name the parent shape**. Next give a perspective drawing of the molecular shape. In the perspective drawing multiple bonds, formal charges on atoms and lone pairs might be omitted for clarity. **Name the molecular shape**. Finally, indicate if the molecule or ion is polar or not. Please refer to the example below (SO_2) if in doubt about any of the instructions.

Lewis Structure	EGG	Parent shape	molecular shape	polar (Yes or No)
	**3**	 trigonal planar	 bent	Yes
SO_2				
	**5**	 trigonal bipyramidal	 Linear	No
	**4**	 tetrahedral	 bent	Yes
	**3**	 trigonal planar	 bent	Yes

2. (**6 marks**) Draw all possible Lewis structures for the carbonate ion. Show lone pairs as pairs of dots and show any non-zero formal charges. Are any of the structures you have drawn equivalent Lewis structures? Circle equivalent Lewis structures.



All three resonance structures are equivalent

3. (**1 marks**) Given the densities of the metals bellow estimate the density of Magnesium. Densities in g/cm^3 at 300K: Na(s) = 0.971, Al(s) = 2.70,

   Since properties change periodically and not abruptly we would expect to observe a trend in densities as we move across the third row elements (at least for the metals which have similar structures). Since Magnesium is between sodium and aluminum an average would be expected:

   d_{Mg} is approximately $(d_{Na} + d_{Al})/2$ = 1.83 g/cm^3 (observed density at 300 K is 1.74)

4. (**7 marks**) Complete the following phrases using the words **increases, decreases, larger, smaller**,

   Ionization energy **decreases** from the top to the bottom in a group.

   Electron affinity **increases** from the left to the right within a period.

   Atomic radius **increases** from the top to the bottom in a group.

   Electronegativity **increases** from the left to the right within a period.

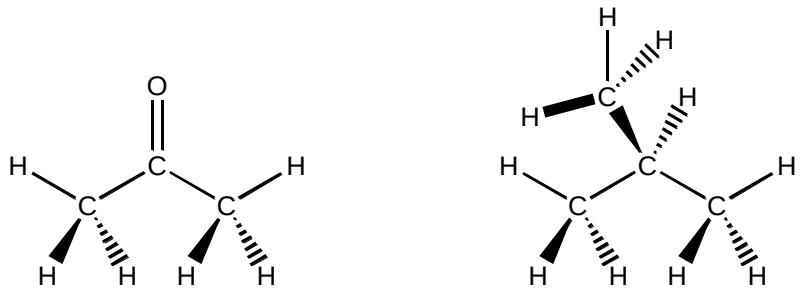
   The metallic character of an element **decreases** from the left to the right within a period.

   Cations are **smaller** than the atoms from which they form.

   In isoelectronic anions the radius **increases** as the charge increases (becomes more negative).

Part 3 Intermolecular interactions, Colligative Properties and Gases and Stoichiometry

1. (**3 marks**) Shown below are the perspective drawings of two simple organic compounds A and B. Both compounds have similar molecular masses but differ hugely in their physical and chemical properties.
   Identify the high and low boiling compound. In one or two sentences give the reason for your choice. Use arguments involving intermolecular interactions.



or simplified

Compound A

High boiling

Compound B

Low boiling

Both compounds have similar mass (number of electrons) and thus similar London dispersion forces but compound A is polar whilst compound B is not. The additional dipole-dipole interactions in compound A will require additional force (heat) to be broken which will make the compound harder to boil. Compound A (acetone) is a liquid at room temperature whilst Compound B (isobutane, the main component in lighters) is a gas at room temperature and 1 atm.

Careful, Hydrogen bonding is not possible between molecules of A since a hydrogen bound to O,N,or F is not present!

2. (**2 marks**) Arrange the following species in order of **increasing** boiling point: S, CO_2, Br_2, N_2.

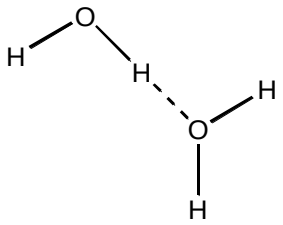
   N_2 < CO_2 < Br_2 <S

   Remember the stoichiometry lab? Sulfur is a yellow solid (actually cyclic S_8)

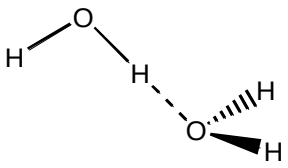
3. (**3 marks**) Decide if the following properties of matter are effected or governed by intermolecular interactions

Surface tension	(**<u>Yes</u>** or No)
Boiling point	(**<u>Yes</u>** or No)
Molar mass	(Yes or **<u>No</u>**)
Solubility	(**<u>Yes</u>** or No)
Vapour pressure	(**<u>Yes</u>** or No)
Conductivity	(Yes or **<u>No</u>**)

4. (**3 marks**) Draw two water molecules hydrogen bonded. Be sure to represent the correct geometry of the molecules in your drawing and align the molecules properly. Draw covalent bonds as solid lines (as usual) and the hydrogen bond as dashed line.



or better:

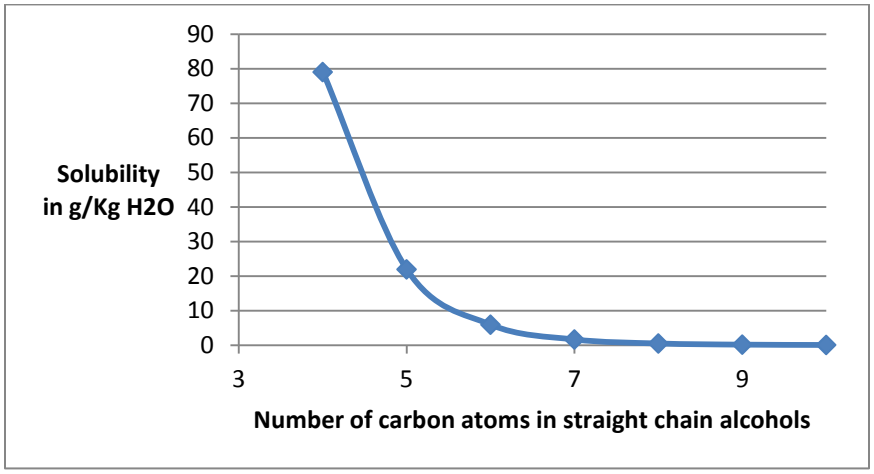


5. (**2 marks**) The solubility of ammonia (NH_3) and nitrogen (N_2) in water differ greatly. Which gas would you expect to be more soluble in water and why?

NH_3 is polar and will be able to dissolve in polar water (engaging in hydrogen bonding as well as ionizing). N_2 on the other hand is non-polar and is not able to hydrogen bond to water (the nitrogen must be polarized $\boldsymbol{\delta^-}$) and will not dissolve well in water. Likes dissolves likes.

6. The graph and table bellow show the aqueous solubility (in g / kg H_2O) of straight chain alcohols as a function of the chain length (see table bellow).

alcohol	# of C-atoms	Aqueous solubility (g/kgH_2O) @ 25°C
CH_3-OH	1	Completely miscible with H_2O
CH_3-CH_2-OH	2	Completely miscible with H_2O
CH_3-CH_2-CH_2-OH	3	Completely miscible with H_2O
CH_3-$(CH_2)_2$-CH_2-OH	4	79.0
CH_3-$(CH_2)_3$-CH_2-OH	5	21.9
CH_3-$(CH_2)_4$-CH_2-OH	6	5.9
CH_3-$(CH_2)_5$-CH_2-OH	7	1.62
CH_3-$(CH_2)_6$-CH_2-OH	8	0.46
CH_3-$(CH_2)_7$-CH_2-OH	9	0.129
CH_3-$(CH_2)_8$-CH_2-OH	10	0.037



a) (**2 marks**) In a few paragraphs explain the observed trend that the aqueous solubility of the straight chain alcohols decreases with chain length. Also explain why the first members (methanol, ethanol and propanol) are completely miscible with water.

Methanol (sometimes referred to as organic water), ethanol and propanol are completely miscible with water because the hydrogen bonding between those alcohols is comparable in strength to the hydrogen bonding in water and London forces in the short alkyl chains are weak. As the chain grows though London forces become stronger and finally dominate intermolecular forces. The heavier alcohols can interact much better with each other than with water and solubility decreases as chain length increases.

b) (**6 marks**) Calculate the vapour pressure of a saturated aqueous solution of butanol ($C_4H_{10}O$) in water at 25°C. The vapour pressure of pure water at that temperature is 3.17 kPa. The vapour pressure of pure butanol is 0.86 kPa.

This is essentially the benzene/toluene problem from the problem set, except the data need to be gathered from a table:

The solubility of butanol is 79.0 g/kg water = 1.066 mol butanol (FW = 74.124 g/mol) in 55.5 mol water. Since this is a mixture of two volatiles we use Raoult's law twice:

$p_{H2O} = X_{H2O}\, p°_{H2O} = \frac{55.5\ mol}{1.066\ mol + 55.5\ mol}$ 3.17 kPa = 3.11 kPa

$p_{Butanol} = X_{Butanol}\, p°_{Butanol} = \frac{1.066\ mol}{1.066\ mol + 55.5\ mol}$ 0.86 kPa = 0.0162 kPa

The vapour pressure over the solution is the sum of the two: $P_{total} = p_{H2O} + p_{Butanol}$ = **3.13 kPa**

c) (**2 marks**) What is the composition of the vapour over the solution?

$\mathbf{X_{Butanol}} = \frac{p(Butanol)}{P(total)} = \frac{0.0162\ \text{kPa}}{3.13\ \text{kPa}} = \mathbf{0.00518}$ (or 0.5 %)

$\mathbf{X_{water}} = \frac{p(water)}{P(total)} = \frac{3.11\ \text{kPa}}{3.13\ \text{kPa}} = \mathbf{0.9948}$ (or 99.5 %)

7. (**5 marks**) Complete combustion of a 1.110 g sample of a gaseous hydrocarbon (e.g. a compound composed only of the elements carbon and hydrogen) yields 3.613 g carbondioxide and 1.109 g water. A 0.288 g sample of the hydrocarbon occupies a volume of 131 mL at 24.8 °C and 753 mm Hg. What is the molecular formula of the hydrocarbon? Propose a valid Lewis structure for the hydrocarbon.

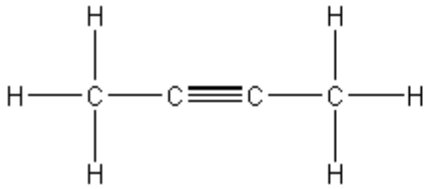
3.613 g CO_2 $\frac{mol}{44.011\ g}$ = 0.08209 mol CO_2 0.08209 mol CO_2 $\frac{1\ mol\ C}{1\ mol\ CO_2}$ = 0.08209 mol C

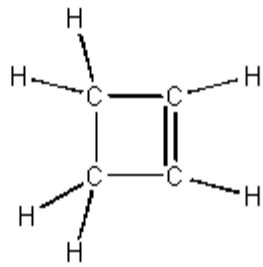
1.109 g H_2O $\frac{mol}{18.016\ g}$ = 0.06156 mol H_2O 0.06156 mol H_2O $\frac{2\ mol\ H}{1\ mol\ H_2O}$ = 0.1231 mol H

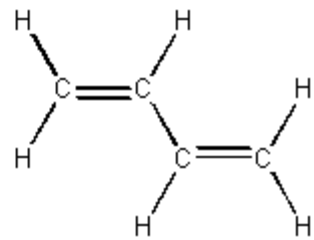
Ratio C:H in the hydrocarbon is 1:1.5 or better 2:3 so we have $(C_2H_3)_n$

pV = nRT = $\frac{mRT}{FW}$ thus FW = $\frac{mRT}{pV}$ = $\frac{(0.288\ g)(8.314\ L\ kPa\ mol^{-1}K^{-1})(298\ K)(760\ mm\ Hg)}{(753\ mm\ Hg)(101.3\ kPa)(0.131\ L)}$ = **54.27 g/mol**

(2 x 12.011 g/mol + 3 x 1.008 g/mol)$_n$ = 54.27 g/mol thus n = 2.006
The hydrocarbon is $\mathbf{C_4H_6}$







Here are a few possible Lewis structures.

8. (**4 marks**) A mixture of 1.25 g helium, 1.30 g carbon dioxide and 1.2 g nitrogen is confined to a 2.5 L flask at room temperature (22 °C). Calculate the partial pressure of each gas and the total pressure of the mixture assuming ideal gas behaviour.

n_{He} = 0.3123 mol, n_{CO2} = 0.02954 mol, n_{N2} = 0.04284 mol **n_{total} = 0.3847**

$\mathbf{p_{total}} = \frac{nRT}{V} = \frac{(0.3847\ mol)(8.314\ L\ kPa\ mol^{-1}K^{-1})(294\ K)}{2.5L}$ = **376.1 kPa**

$\mathbf{P_{He}} = \frac{0.3123\ mol}{0.3847\ mol}$ 376.1 kPa = 305 kPa

$\mathbf{P_{CO2}} = \frac{0.0295\ mol}{0.3847\ mol}$ 376.1 kPa = 28.9 kPa

$\mathbf{P_{N2}} = \frac{0.04284\ mol}{0.3847\ mol}$ 376.1 kPa = 41.9 kPa

9. (**5 marks**) At 25 ˚C and a CO_2 pressure of 1 atm 0.0310 mol of CO_2 dissolve in enough water to make 1 L of solution. What is the molar solubility of CO_2 if its pressure is raised to 5.3 atm?

This is a Henry's law problem: $S_g = kp_g$ first we find k, then S_{CO2} at 5.3 atm

$k = \frac{0.0310\ mo\ L^{-1}l}{1\ atm}$ = 0.0310 mol L^{-1} atm^{-1},

$S_{CO2} = kp_{CO2}$ = (0.0310 mol L^{-1} atm^{-1})(5.3 atm) = 0.1643 mol/L = **0.164 M**

10. (**8 marks**) Hydrogen can be produced by reacting silicon with sodium hydroxide according to the following equation.

$$\mathbf{Si(s) + 4NaOH(aq) \rightarrow Na_4SiO_4(aq) + 2H_2(g)}$$

Calculate the volume of Hydrogen that can be produced at 1.02 atm and 22° C if 5.97 g of silicon are reacted with 20.3 g sodium hydroxide in water.

First we find the limiting reactant:

5.97 g Si $\frac{mol}{28.08\ g}$ = 0.2126 mol Si, 0.2126 mol Si $\frac{2\ mol\ H_2}{1\ mol\ Si}$ = 0.425 mol H_2

20.3 g NaOH $\frac{mol}{40.0\ g}$ = 0.5075 mol NaOH, 0.5075 mol NaOH $\frac{2\ mol\ H_2}{4\ mol\ NaOH}$ = 0.254 mol H_2

Sodium hydroxide is the limiting reactant. It limits the amount of hydrogen that can form.

$$V = \frac{nRT}{p} = \frac{(0.254\ \text{mol H}_2)(8.314\ L\ kPa\ mol^{-1}K^{-1})(295\ K)}{(1.02\ atm)(101.3\ kPa\ atm^{-1})} = \mathbf{6.02\ L}$$

This seems reasonable because at STP we would have (22.4 L mol^{-1})(0.254 mol) = 5.69 L

11. (**6 marks**) 1.05 mol of carbon dioxide occupy a vessel of 2.50 L at -15°C. Calculate the pressure of the gas using a) the ideal gas equation and b) the van der Waals equation. The van der Waals constants for carbon dioxide are a = 3.62 x10^2 L^2kPa/mol^2 and b = 0.0427 L/mol.

a) $p = \frac{nRT}{V} = \frac{(1.05\ mol)(8.314\ kPa\ L\ mol^{-1}K^{-1})(258\ K)}{2.5\ L}$ = 900.9 kPa $\mathbf{p_{ideal} = 901\ kPa}$

b) $p = \frac{nRT}{V-nb} - \frac{n^2a}{V^2} = \frac{(1.05\ mol)(8.314\ kPa\ L\ mol^{-1}K^{-1})(258\ K)}{2.5\ L-(1.05\ mol)(0.0427\ l\ mol^{-1})} - \frac{(1.05\ mol)^2(362\ L^2\ kPa\ mol^{-2})}{(2.5\ L)^2}$

p = 917.36 kPa – 63.86 kPa $\mathbf{p_{real} = 853\ kPa}$